

Smoking Pipe CFD Simulation

OpenFOAM v2012 · Salome ·
ParaView
Three simulations: Steady-State ·
Transient · Breathing Cycle

SpaceClaim

Salome

OpenFOAM v2012

snappyHexMesh

simpleFoam

pimpleFoam

ParaView

Project Overview

This project simulates airflow through a smoking pipe geometry using three progressively complex OpenFOAM setups: a steady-state baseline, a transient impulsive start, and a physiologically realistic breathing cycle with a coded time-varying boundary condition. All tools are open-source.

Full workflow documented at: github.com/samuelsimmons99/OpenFOAM-Portfolio — Smoking Pipe Tutorial

Mesh (shared across all simulations)

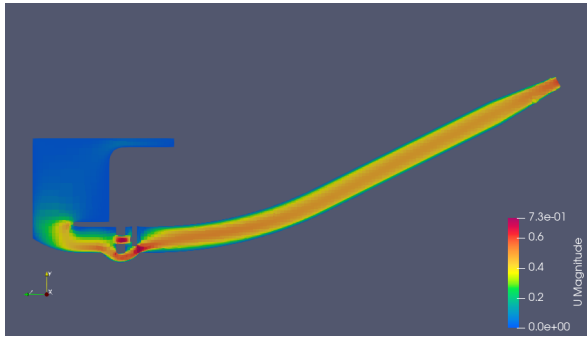
Metric	Value	
Total cells	807,523	✓
Hexahedra	739,338 (92%)	✓
Max non-orthogonality	53.4°	✓
Max skewness	2.32	✓
Mesh checks failed	0	✓

Simulation 1 — Steady-State Flow

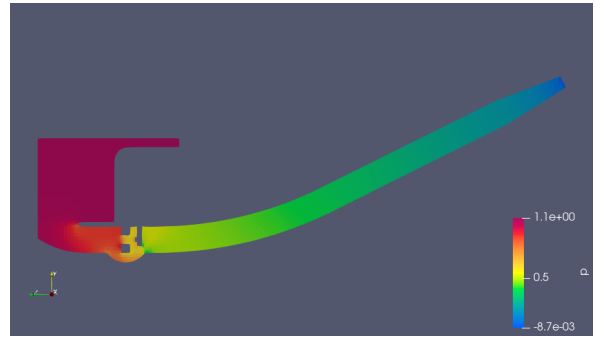
simpleFoam · SIMPLE
algorithm

Setup: Fixed inlet velocity, no-slip walls, pressure outlet. Laminar Stokes model. Converged in **51** iterations.

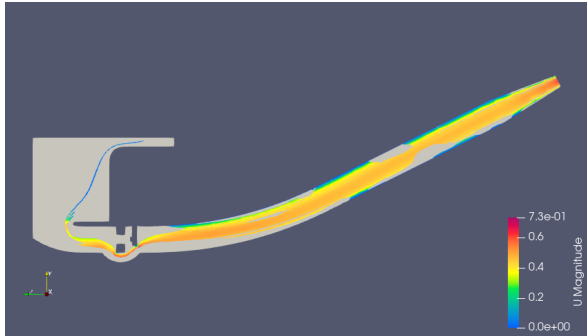
Patch	U	p
Inlet	fixedValue (0 0 -0.5 m/s)	zeroGradient
Outlet	inletOutlet	fixedValue 0
Walls / PipeHandle	noSlip	zeroGradient



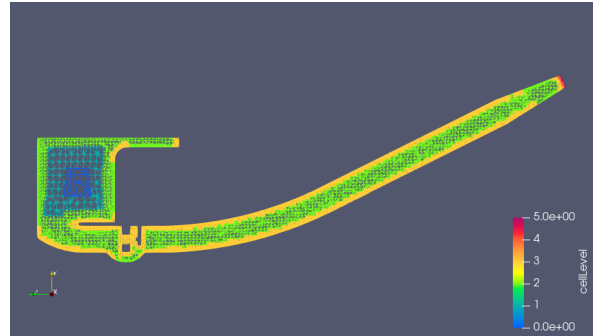
Velocity magnitude — centreline slice



Pressure distribution — inlet to outlet



Streamlines coloured by U magnitude



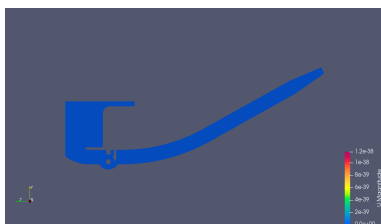
Mesh refinement levels

Simulation 2 — Transient Impulsive Start

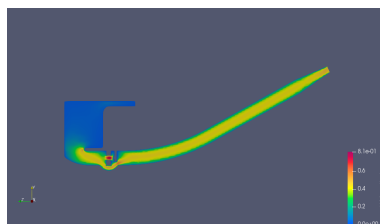
pimpleFoam · PIMPLE
algorithm

Setup: Same geometry and boundary conditions as steady-state but started from rest (zero internal field) using pimpleFoam. Adaptive Co-based timestepping with $\maxCo=0.5$. Flow reached steady state by $t=0.25s$.

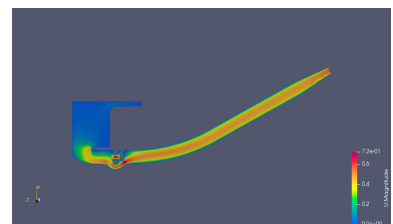
Parameter	Value
Solver	pimpleFoam
nOuterCorrectors	2
nCorrectors	2
nNonOrthogonalCorrectors	3
maxCo	0.5
Simulation time	0.5s



$t=0s$ — zero velocity, flow at rest



$t=0.01s$ — impulsive start, jet forming



$t=0.1s$ — approaching steady state

Simulation 3 — Physiological Breathing Cycle

pimpleFoam ·
codedFixedValue BC

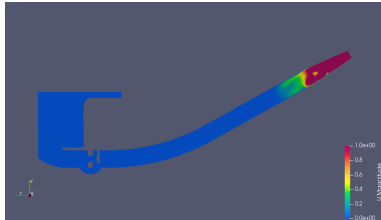
Setup: Time-varying suction boundary condition at the outlet (mouthpiece) using OpenFOAM's *codedFixedValue* to simulate realistic breathing. Flow is driven by suction rather than a fixed inlet velocity. Simulation ran for **8 seconds** capturing two complete breath cycles.

Breathing parameters:

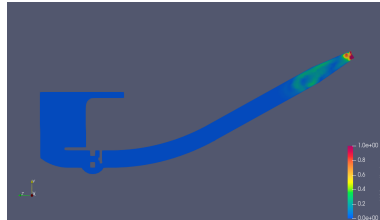
```
breathingStart = 0.1s // first breath begins  
  
tB = 1.5s // breath duration  
  
nextBreath = 5.0s // pause between breaths  
  
dotVBreath = 0.002/1.5 = 1.33e-3 m³/s // flow rate
```

Breathing cycle timeline:

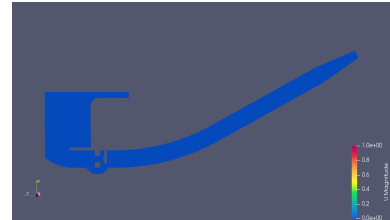
Time	Phase	Flow
0.0 → 0.1s	Pre-breath	No flow
0.1 → 1.6s	First breath	Suction active — flow drawn through pipe
1.6 → 6.6s	Pause	Flow decays — pipe clears
6.6 → 8.1s	Second breath	Suction resumes



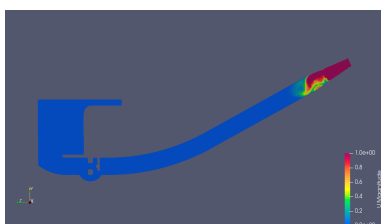
t=1s — active breath, suction jet at mouthpiece



t=2s — end of first breath, flow decelerating



t=4s — pause phase, near-zero flow



t=7s — second breath, suction jet re-established

Key Challenges & Solutions

Tetrahedral background mesh

snappyHexMesh requires hex cells. Rebuilt using Salome hex algorithm and ideasUnvToFoam.

Open STL edges

136 open edges at outlet caused snappyHexMesh to hang. Fixed by re-sewing geometry in Salome.

Persistent bad mesh face

Single 161° non-orthogonal face located at (-0.009, 0.017, -0.020) via face index lookup. Eliminated using a searchableSphere refinement region targeting that cell.

Adaptive timestep collapse

deltaT collapsed to $\sim 1e-62$ due to bad face producing spurious Co spike. Fixed by minDeltaT and switching writeControl to timeStep.

codedFixedValue timing

Breathing condition used absolute simulation time — required offsetting breathingStart when restarting from non-zero time.